

Setting up matplotlibcpp on Mac:

1. Download the following three files from the week 12 module on bCourses:
 - a. **matplotlibcpp.h** (this version is modified so redownload it even if you downloaded it in class)
 - b. **Makefile**
 - c. **12_lab_demo.cpp**
2. Put all three files in the same folder and open a VS code window to that folder.
3. In the terminal, deactivate conda using 'conda deactivate'
4. Run the following: 'brew doctor'
 - a. If you receive the message 'Your system is ready to brew', move onto step 5.
 - b. If you receive an error, complete the following steps:
 - i. Run this command in the terminal:

```
/bin/bash -c "$(curl -fsSL  
https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```
 - ii. After the download is complete you will receive a message that says ==> Next Steps.
Copy and run the provided commands.
 - iii. Rerun 'brew doctor' to make sure the download worked correctly.
5. Run: 'brew install python numpy python-matplotlib'
6. Run the command 'make'.

Warnings may pop up, but your file should compile without any errors. You will see a message saying 'Build complete. 'Run with './12_lab_demo' if the compilation was successful.'

7. You should see an executable called '12_lab_demo' appear in your folder. Run this using './12_lab_demo'. A graph should pop up when you run this command.
8. Update the Makefile to compile different cpp files by changing the SRCS and TARGET variables at the top of the file.
9. In order to recompile your cpp file, the executable must be removed from the folder. You can do this in two ways:
 - a. Manually delete the executable from the folder.
 - b. Run the command 'make clean'.

Once the file is removed, run 'make' again to recompile.

Note: Make sure 'matplotlibcpp.h' and the Makefile are in the same directory as your source cpp file to correctly use the make command for your final project.